Food-borne Pathogens and Molecular Surveillance

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Food-borne Pathogens and Molecular Surveillance (A11406)

Short Title:	Food-borne Pathogen & Mol Surv		
Faculty	Science and Computing		
Department	Science		
Cluster	Food Science		
Author	ROMAHONY		
Credits	5	**Level:**	Intermediate

Description of Module / Aims

Students will gain a good knowledge of current and emerging food-borne pathogens, and will understand the importance of their impact on public health. Students will be informed of modern diagnostic tools for molecular surveillance of food-borne pathogens, as well as relevant control programmes.

Programmes

`HEAL-0079`	BA (Hons) in Health Promotion (`WD_HHPRM_B`)	4 / 7 / M
`FOOD-0022`	BSc (Hons) in Food Science and Innovation (`WD_SFSIN_B`)	3 / 6 / M
`FOOD-0022`	BSc in Food Science with Business (`WD_SFDSC_D`)	3 / 6 / M

Indicative Content

- Current bacterial foodborne pathogens such as Escherichia coli O157:H7 and other emerging verotoxigenic E. coli, Salmonella spp., Campylobacter spp., Listeria spp., Cronobacter sakazakii
- Non-bacterial food pathogens, e.g. norovirus, rotavirus and cryptosporidium
- Multi drug resistant (MDR) bacteria, including the emergence of fluoroquinolone resistance in important food-borne pathogens such as Campylobacter spp. and Salmonella spp.
- Control of zoonotic pathogens and relevant control programmes, e.g. Salmonella control in pigs
- Traditional and molecular approaches to food safety monitoring: this will include bacteriology, molecular detection and molecular surveillance tools

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Explore the various current and emerging pathogens.
2. Distinguish between food-borne pathogens i.e. bacterial and non-bacterial (viral and parasitic).
3. Investigate microbial drug resistance (MDR) and highlight the concerns of MDR in relation to public health.
4. Distinguish between a variety of modern molecular tools used for pathogen surveillance.
5. Debate the need for molecular surveillance tools in relation to food-borne pathogens.
6. Examine national food-borne pathogen control programmes, e.g. Salmonella Control Programme.

Learning and Teaching Methods

- Lectures -- blended learning.
- Formative learning using case studies, e.g. FSAI alert reports.
- Flipped classroom using e-activities.
- In-class presentations.
- Self-study of assigned reading.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	60%	2,3,4,6
Continuous Assessment	40%	
Assignment	40%	1,5

Assessment Criteria

<40%: No understanding of the basics of food-borne pathogens.
40%-49%: A basic understanding of food-borne pathogens and their relevance.
50%-59%: Understands the importance of zoonoses and the necessity for international surveillance and control.
60%-69%: Can critically evaluate modern diagnostic tools for tracing food pathogens.
70%-100%: Can perform all the previous assessment criteria. Can apply that knowledge in a global context, and evaluate the relevance from a public health perspective.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Tutorial	12	
Independent Learning	99	

Supplementary Material

- "Department of Agriculture Food & the Marine ." www.agriculture.gov.ie
- "European Food Safety Authority ." www.efsa.europa.eu
- "Food Safety Authority of Ireland (www.fsai.ie)." www.fasi.ie

Requested Resources

- EQUIPMENT: MAC PCs
- EQUIPMENT: Blackboard
- Lecture Room: Loose Seated